

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

Slide Set 3: Naïve Bayes



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
 - ▶ For use only by enrolled students in the course

A toy problem

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

A toy problem

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

A toy problem

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

$x_2 = \text{cloudy} \rightarrow y = \text{???}$

A toy problem

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

$x_2 = \text{cloudy} \rightarrow y = \text{????}$

$x_2 = \text{raining} \rightarrow y = \text{????}$

$x_2 = \text{sunny} \rightarrow y = \text{????}$

A toy problem

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

$x_2 = \text{cloudy} \rightarrow y = \text{Rte-75}$

$x_2 = \text{raining} \rightarrow y = \text{????}$

$x_2 = \text{sunny} \rightarrow y = \text{Rte-113}$

counts	Rte-113	Rte-75
cloudy	1	2
raining	1	1
sunny	4	1

A toy problem

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

$x_2 = \text{cloudy} \rightarrow y = \text{Rte-75}$

$x_2 = \text{raining} \rightarrow y = \text{????}$

$x_2 = \text{sunny} \rightarrow y = \text{Rte-113}$

counts	Rte-113	Rte-75
cloudy	1	2
raining	1	1
sunny	4	1

$\hat{\text{Pr}}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

Bayes' rule

$\hat{\Pr}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

Bayes' rule

$\hat{\Pr}(x_2, y)$	Rte-113	Rte-75	$\hat{\Pr}(x_2)$
cloudy	0.1	0.2	0.3
raining	0.1	0.1	0.2
sunny	0.4	0.1	0.5

Bayes' rule

$\hat{Pr}(x_2, y)$	Rte-113	Rte-75	$\hat{Pr}(x_2)$
cloudy	0.1	0.2	0.3
raining	0.1	0.1	0.2
sunny	0.4	0.1	0.5

$\hat{Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

Bayes' rule

$\hat{\Pr}(x_2, y)$	Rte-113	Rte-75	$\hat{\Pr}(x_2)$	$\hat{\Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.1	0.2	cloudy	0.33	0.33	0.67
raining	0.1	0.1	raining	0.50	0.50	0.50
sunny	0.4	0.1	sunny	0.80	0.80	0.20

$\hat{\Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

Bayes' rule

$\hat{\Pr}(x_2, y)$	Rte-113	Rte-75	$\hat{\Pr}(x_2)$	$\hat{\Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.1	0.2	cloudy	0.33	0.67	
raining	0.1	0.1	raining	0.50	0.50	
sunny	0.4	0.1	sunny	0.80	0.20	

$\hat{\Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$\hat{\Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

Bayes' rule

$\hat{Pr}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

$\hat{\Pr}(x_2)$	
cloudy	0.3
raining	0.2
sunny	0.5

$\hat{Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.33	0.67
raining	0.50	0.50
sunny	0.80	0.20

$\hat{Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$$P(A=a) = \sum_b P(A=a, B=b)$$

$\hat{Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

Bayes' rule

$\hat{Pr}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

$\hat{Pr}(x_2)$		$\hat{Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.3	cloudy	0.33	0.67
raining	0.2	raining	0.50	0.50
sunny	0.5	sunny	0.80	0.20

$\hat{Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$$P(A=a) = \sum_b P(A=a, B=b)$$

$\hat{Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

$$\begin{aligned} P(A=a, B=b) &= P(A=a)P(B=b | A=a) \\ &= P(B=b)P(A=a | B=b) \end{aligned}$$

Bayes' rule

$\hat{Pr}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

$\hat{Pr}(x_2)$		$\hat{Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.3	cloudy	0.33	0.67
raining	0.2	raining	0.50	0.50
sunny	0.5	sunny	0.80	0.20

$\hat{Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$$P(A=a) = \sum_b P(A=a, B=b)$$

$\hat{Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

$$\begin{aligned} P(A=a, B=b) &= P(A=a)P(B=b | A=a) \\ &= P(B=b)P(A=a | B=b) \end{aligned}$$

$$P(A=a | B=b) = \frac{P(A=a, B=b)}{P(B=b)}$$

Bayes' rule

$\hat{Pr}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

$\hat{Pr}(x_2)$		$\hat{Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.3	cloudy	0.33	0.67
raining	0.2	raining	0.50	0.50
sunny	0.5	sunny	0.80	0.20

$\hat{Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$$P(A=a) = \sum_b P(A=a, B=b)$$

$\hat{Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

$$\begin{aligned} P(A=a, B=b) &= P(A=a)P(B=b | A=a) \\ &= P(B=b)P(A=a | B=b) \end{aligned}$$

$$P(A=a | B=b) = \frac{P(A=a, B=b)}{P(B=b)}$$

$$P(A=a | B=b) = \frac{P(A=a, B=b)}{P(B=b)}$$

Bayes' rule

$\hat{\Pr}(x_2, y)$	Rte-113	Rte-75
cloudy	0.1	0.2
raining	0.1	0.1
sunny	0.4	0.1

$\hat{\Pr}(x_2)$		$\hat{\Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.3	cloudy	0.33	0.67
raining	0.2	raining	0.50	0.50
sunny	0.5	sunny	0.80	0.20

$\hat{\Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$\hat{\Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

$$P(A=a) = \sum_b P(A=a, B=b)$$

$$\begin{aligned} P(A=a, B=b) &= P(A=a)P(B=b | A=a) \\ &= P(B=b)P(A=a | B=b) \end{aligned}$$

$$P(A=a | B=b) = \frac{P(A=a, B=b)}{P(B=b)}$$

$$P(A=a | B=b) = \frac{P(B=b|A=a)P(A=a)}{P(B=b)} \quad (\text{Bayes' Theorem})$$

Bayes' rule

$\hat{\Pr}(x_2, y)$	Rte-113	Rte-75	$\hat{\Pr}(x_2)$	$\hat{\Pr}(y x_2)$	Rte-113	Rte-75
cloudy	0.1	0.2	cloudy	0.33	0.67	
raining	0.1	0.1	raining	0.50	0.50	
sunny	0.4	0.1	sunny	0.80	0.20	

$\hat{\Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$$P(A=a) = \sum_b P(A=a, B=b)$$

$\hat{\Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

$$\begin{aligned} P(A=a, B=b) &= P(A=a)P(B=b | A=a) \\ &= P(B=b)P(A=a | B=b) \end{aligned}$$

$$P(A=a | B=b) = \frac{P(A=a, B=b)}{P(B=b)}$$

$$P(A=a | B=b) = \frac{P(B=b|A=a)P(A=a)}{P(B=b)} \quad (\text{Bayes' Theorem})$$

$$\text{Example: } \hat{\Pr}(\text{Rte-113} | \text{sunny}) = \frac{\hat{\Pr}(\text{sunny}|\text{Rte-113})\hat{\Pr}(\text{Rte-113})}{\hat{\Pr}(\text{sunny})} = \frac{0.67 \times 0.6}{0.5} = 0.8$$

Approximate Bayes-optimal classifier

$(x_1 = \text{no}, x_2 = \text{sunny}, x_3 = \text{no}) \rightarrow y = \text{???}$

Approximate Bayes-optimal classifier

$(x_1 = \text{no}, x_2 = \text{sunny}, x_3 = \text{no}) \rightarrow y = \text{???}$

Bayes-optimal classifier: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c P(y = c \mid \vec{x})$

Approximate Bayes-optimal classifier

$(x_1 = \text{no}, x_2 = \text{sunny}, x_3 = \text{no}) \rightarrow y = \text{???}$

Bayes-optimal classifier: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c P(y = c \mid \vec{x})$

Using estimated probabilities: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c \hat{Pr}(y = c \mid \vec{x})$

Approximate Bayes-optimal classifier

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

$(x_1 = \text{no}, x_2 = \text{sunny}, x_3 = \text{no}) \rightarrow y = \text{???}$

Bayes-optimal classifier: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c P(y = c \mid \vec{x})$

Using estimated probabilities: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c \hat{Pr}(y = c \mid \vec{x})$

Approximate Bayes-optimal classifier

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)	$\hat{Pr}(x,y)$	Rte-113	Rte-75
no	sunny	no	Rte-113	(no,cloudy,no)	0.1	0.0
no	sunny	yes	Rte-113	(no,cloudy,yes)	0.0	0.1
no	cloudy	yes	Rte-75	(no,raining,no)	0.1	0.0
yes	sunny	no	Rte-113	(no,raining,yes)	0.0	0.1
no	raining	no	Rte-113	(no,sunny,no)	0.2	0.0
no	raining	yes	Rte-75	(no,sunny,yes)	0.1	0.0
yes	cloudy	yes	Rte-75	(yes,cloudy,no)	0.0	0.0
no	sunny	no	Rte-113	(yes,cloudy,yes)	0.0	0.1
yes	sunny	no	Rte-75	(yes,raining,no)	0.0	0.0
no	cloudy	no	Rte-113	(yes,raining,yes)	0.0	0.0
				(yes,sunny,no)	0.1	0.1
				(yes,sunny,yes)	0.0	0.0

$(x_1 = \text{no}, x_2 = \text{sunny}, x_3 = \text{no}) \rightarrow y = \text{????}$

Bayes-optimal classifier: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c P(y = c \mid \vec{x})$

Using estimated probabilities: Given $\vec{x} = (x_1, x_2, \dots, x_n)$, predict $y = \arg \max_c \hat{Pr}(y = c \mid \vec{x})$

Naïve Bayes

Assume: $P(x_1, x_2, \dots, x_n \mid y) = \prod_{i=1}^n P(x_i \mid y)$

$[X_i \text{ is independent of } X_j, \text{ given } Y]$

Naïve Bayes

Assume: $P(x_1, x_2, \dots, x_n \mid y) = \prod_{i=1}^n P(x_i \mid y)$

$[X_i \text{ is independent of } X_j, \text{ given } Y]$

Then use $\hat{P}r(x_i \mid y)$ instead of $P(x_i \mid y)$.

Naïve Bayes

Assume: $P(x_1, x_2, \dots, x_n \mid y) = \prod_{i=1}^n P(x_i \mid y)$

$[X_i \text{ is independent of } X_j, \text{ given } Y]$

Then use $\hat{Pr}(x_i \mid y)$ instead of $P(x_i \mid y)$.

$$\begin{aligned}\hat{Pr}(y \mid \vec{x}) &= \frac{\hat{Pr}(\vec{x} \mid y) \hat{Pr}(y)}{\hat{Pr}(\vec{x})} \\ &\approx \frac{\hat{Pr}(y) \prod_{i=1}^n \hat{Pr}(x_i \mid y)}{\hat{Pr}(\vec{x})}\end{aligned}$$

Naïve Bayes

Assume: $P(x_1, x_2, \dots, x_n \mid y) = \prod_{i=1}^n P(x_i \mid y)$

$[X_i \text{ is independent of } X_j, \text{ given } Y]$

Then use $\hat{Pr}(x_i \mid y)$ instead of $P(x_i \mid y)$.

$$\begin{aligned}\hat{Pr}(y \mid \vec{x}) &= \frac{\hat{Pr}(\vec{x} \mid y) \hat{Pr}(y)}{\hat{Pr}(\vec{x})} \\ &\approx \frac{\hat{Pr}(y) \prod_{i=1}^n \hat{Pr}(x_i \mid y)}{\hat{Pr}(\vec{x})}\end{aligned}$$

Naïve Bayes rule:

Given $\vec{x} = (x_1, x_2, \dots, x_n)$,

Pick c that maximizes $\hat{Pr}(y=c) \prod_{i=1}^n \hat{Pr}(x_i \mid y=c)$

Naïve Bayes Example

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

Naïve Bayes Example

x_1 (weekend?)	x_2 (weather)	x_3 (game?)	y (faster route)
no	sunny	no	Rte-113
no	sunny	yes	Rte-113
no	cloudy	yes	Rte-75
yes	sunny	no	Rte-113
no	raining	no	Rte-113
no	raining	yes	Rte-75
yes	cloudy	yes	Rte-75
no	sunny	no	Rte-113
yes	sunny	no	Rte-75
no	cloudy	no	Rte-113

$$\hat{\Pr}(y) \quad \begin{matrix} \text{Rte-113} & \text{Rte-75} \\ \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{matrix}$$

$$\hat{\Pr}(x_1|y) \quad \begin{matrix} \text{Rte-113} & \text{Rte-75} \\ \text{no} \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{matrix}$$

$$\hat{\Pr}(x_2|y) \quad \begin{matrix} \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{matrix}$$

$$\hat{\Pr}(x_3|y) \quad \begin{matrix} \text{Rte-113} & \text{Rte-75} \\ \text{no} \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{matrix}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

$$\hat{\Pr}(y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\hat{\Pr}(x_1|y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \text{no} \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\hat{\Pr}(x_2|y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\hat{\Pr}(x_3|y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \text{no} \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\hat{\Pr}(y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\hat{\Pr}(x_1|y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \begin{array}{c} \text{no} \\ \text{yes} \end{array} \left[\begin{array}{cc} 0.83 & 0.50 \\ 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\hat{\Pr}(x_2|y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \begin{array}{c} \text{cloudy} \\ \text{raining} \\ \text{sunny} \end{array} \left[\begin{array}{cc} 0.17 & 0.50 \\ 0.17 & 0.25 \\ 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\hat{\Pr}(x_3|y) \quad \begin{array}{cc} \text{Rte-113} & \text{Rte-75} \\ \begin{array}{c} \text{no} \\ \text{yes} \end{array} \left[\begin{array}{cc} 0.83 & 0.25 \\ 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$\hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-113})$

$\hat{\Pr}(y)$	Rte-113	Rte-75
	0.6	0.4

$\hat{\Pr}(x_1 y)$	Rte-113	Rte-75
no	0.83	0.50
yes	0.17	0.50

$\hat{\Pr}(x_2 y)$	Rte-113	Rte-75
cloudy	0.17	0.50
raining	0.17	0.25
sunny	0.67	0.25

$\hat{\Pr}(x_3 y)$	Rte-113	Rte-75
no	0.83	0.25
yes	0.17	0.75

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.83 \times 0.17 \times 0.83 = 0.0703 \end{aligned}$$

$$\begin{array}{cc} \hat{\Pr}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.83 \times 0.17 \times 0.83 = 0.0703 \end{aligned}$$

Possibility: $y = \text{Rte-75}$

$$\begin{array}{cc} \hat{\Pr}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.83 \times 0.17 \times 0.83 = 0.0703 \end{aligned}$$

Possibility: $y = \text{Rte-75}$

$$\hat{\Pr}(y=\text{Rte-75})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-75})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-75})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-75})$$

$$\begin{array}{cc} \hat{\Pr}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.83 \times 0.17 \times 0.83 = 0.0703 \end{aligned}$$

Possibility: $y = \text{Rte-75}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-75})\hat{\Pr}(x_1=\text{no} \mid y=\text{Rte-75})\hat{\Pr}(x_2=\text{cloudy} \mid y=\text{Rte-75})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-75}) \\ &= 0.4 \times 0.50 \times 0.50 \times 0.25 = 0.025 \end{aligned}$$

$$\begin{array}{cc} \hat{\Pr}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example

Input: $\vec{x} = (\text{no}, \text{cloudy}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\text{Pr}}(y=\text{Rte-113})\hat{\text{Pr}}(x_1=\text{no} \mid y=\text{Rte-113})\hat{\text{Pr}}(x_2=\text{cloudy} \mid y=\text{Rte-113})\hat{\text{Pr}}(x_3=\text{no} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.83 \times 0.17 \times 0.83 = 0.0703 \end{aligned}$$

Possibility: $y = \text{Rte-75}$

$$\begin{aligned} & \hat{\text{Pr}}(y=\text{Rte-75})\hat{\text{Pr}}(x_1=\text{no} \mid y=\text{Rte-75})\hat{\text{Pr}}(x_2=\text{cloudy} \mid y=\text{Rte-75})\hat{\text{Pr}}(x_3=\text{no} \mid y=\text{Rte-75}) \\ &= 0.4 \times 0.50 \times 0.50 \times 0.25 = 0.025 \end{aligned}$$

Therefore, the rule reports Rte-113

$$\begin{array}{cc} \hat{\text{Pr}}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\text{Pr}}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\text{Pr}}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\text{Pr}}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example II

Input: $\vec{x} = (\text{yes}, \text{raining}, \text{no})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{yes} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{raining} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.17 \times 0.17 \times 0.83 = 0.0139 \end{aligned}$$

Possibility: $y = \text{Rte-75}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-75})\hat{\Pr}(x_1=\text{yes} \mid y=\text{Rte-75})\hat{\Pr}(x_2=\text{raining} \mid y=\text{Rte-75})\hat{\Pr}(x_3=\text{no} \mid y=\text{Rte-75}) \\ &= 0.4 \times 0.50 \times 0.25 \times 0.25 = 0.0125 \end{aligned}$$

Therefore, the rule reports Rte-113

$$\begin{array}{cc} \hat{\Pr}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$

Naïve Bayes Example III

Input: $\vec{x} = (\text{yes}, \text{sunny}, \text{yes})$

Possibility: $y = \text{Rte-113}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-113})\hat{\Pr}(x_1=\text{yes} \mid y=\text{Rte-113})\hat{\Pr}(x_2=\text{sunny} \mid y=\text{Rte-113})\hat{\Pr}(x_3=\text{yes} \mid y=\text{Rte-113}) \\ &= 0.6 \times 0.17 \times 0.67 \times 0.17 = 0.0111 \end{aligned}$$

Possibility: $y = \text{Rte-75}$

$$\begin{aligned} & \hat{\Pr}(y=\text{Rte-75})\hat{\Pr}(x_1=\text{yes} \mid y=\text{Rte-75})\hat{\Pr}(x_2=\text{sunny} \mid y=\text{Rte-75})\hat{\Pr}(x_3=\text{yes} \mid y=\text{Rte-75}) \\ &= 0.4 \times 0.50 \times 0.25 \times 0.75 = 0.0375 \end{aligned}$$

Therefore, the rule reports Rte-75

$$\begin{array}{cc} \hat{\Pr}(y) & \text{Rte-113} & \text{Rte-75} \\ & \left[\begin{array}{cc} 0.6 & 0.4 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_1|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.50 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_2|y) & \text{Rte-113} & \text{Rte-75} \\ \text{cloudy} & \left[\begin{array}{cc} 0.17 & 0.50 \end{array} \right] \\ \text{raining} & \left[\begin{array}{cc} 0.17 & 0.25 \end{array} \right] \\ \text{sunny} & \left[\begin{array}{cc} 0.67 & 0.25 \end{array} \right] \end{array}$$

$$\begin{array}{cc} \hat{\Pr}(x_3|y) & \text{Rte-113} & \text{Rte-75} \\ \text{no} & \left[\begin{array}{cc} 0.83 & 0.25 \end{array} \right] \\ \text{yes} & \left[\begin{array}{cc} 0.17 & 0.75 \end{array} \right] \end{array}$$